

Amazon Linux'te Nginx Üzerine Let's Encrypt SSL Sertifikası Kurulumu



🌐 SSL Nedir?

SSL (Secure Sockets Layer), bir web tarayıcısı ile sunucu arasındaki iletişimi şifreleyen güvenlik protokolüdür.

- Tarayıcı ile sunucu arasındaki tüm veri **şifreli** taşınır.
- Kullanıcıların kişisel verileri, şifreleri, kredi kartı bilgileri korunur.
- Tarayıcı adres çubuğunda görünen  **kilit simgesi** güvenliğin işaretidir.
- Google ve diğer arama motorları HTTPS sitelere SEO avantajı sağlar.

🔑 Let's Encrypt Nedir?

Let's Encrypt, ücretsiz ve açık sertifika otoritesidir (CA).

- Web siteleri için otomatik ve ücretsiz **SSL/TLS sertifikası** sağlar.
- Manuel işlem yerine **Certbot** aracıyla tek komutla kurulum yapılabilir.
- Sertifikalar **90 günde bir** yenilenir (auto-renew ile otomatikleştirilebilir).

👉 Kısacası, Let's Encrypt sayesinde herhangi bir maliyet olmadan web sitenizi güvenli hale getirebilirsiniz.

🎯 Bu Lab'da Ne Yapacağız?

- Amazon Linux 2023 üzerinde **Nginx Web Server** kuracağız.
- **Let's Encrypt (Certbot)** kullanarak ücretsiz SSL sertifikası ekleyeceğiz.
- Cron veya systemd timer ile otomatik yenileme ayarlayacağız.
- HTTPS'i tarayıcıda test edeceğiz.

1 EC2 Kurulumu

- **Services** → **EC2** → **Instances** → **Launch instances.**
- **Name and tags:**
 - **Name:** **Nginx_Web**
- **Application and OS Images (AMI):**
 - **Quick Start** → **Amazon Linux 2023 (kernel-6.1)**

[EC2](#) > [Instances](#) > Launch an instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

My AMIs

Quick Start

 Amazon Linux

 macOS

 Ubuntu

 Windows

 Red Hat

 SUSE Linux

 Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-0b09ffb6d8b58ca91 (64-bit (x86), uefi-preferred) / ami-0b2f2a5d7a4d7f37f (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250908.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-0b09ffb6d8b58ca91	2025-09-06	ec2-user

[Verified provider](#)

- **Instance type:** **t2.micro**

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- **Key Pair (login):**
 - **Create new key pair** → **Key pair name:** `ec2_ssh_key`
 - **Type:** RSA, **File format:** `.pem` → **Create** ve seç.

Create key pair [X]

Key pair name
Key pairs allow you to connect to your instance securely.

`ec2_ssh_key`

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#) [↗]

Cancel **Create key pair**

Key dosyasını bilgisayarına kaydet

- **Network settings** → **Edit:**
 - **VPC:** `Default`
 - **Subnet:** varsayılan
 - **Auto-assign public IP:** `Enable`
 - **Firewall (SG):** **Create a new security group**
 - **Security group name:** `Nginx-Web_SG`
 - **Description:** `Security group for Nginx-Web_SG`

- **Rules:**

- **SSH** → **Source:** Anywhere (0.0.0.0/0)
- **HTTP** → **Source:** Anywhere (0.0.0.0/0)
- **HTTPS** → **Source:** Anywhere (0.0.0.0/0)

▼ Network settings Info

VPC - required Info

vpc-011afb7e7a633837c (default) ↕

Subnet Info

No preference ▾ ↕ Create new subnet

Availability Zone Info

No preference ▾ ↕ Enable additional zones

Auto-assign public IP Info

Enable ▾

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group
 ☐ Select existing security group

Security group name - required

Nglnx-Web_SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-:/0#,@[]+=&{}!\$*

Description - required Info

Security group for Nglnx-Web_SG

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0) Remove

Type Info

ssh ▾

Protocol Info

TCP

Port range Info

22

Source type Info

Anywhere ▾

Source Info

Description - optional Info

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0) Remove

Type Info

HTTP ▾

Protocol Info

TCP

Port range Info

80

Source type Info

Custom ▾

Source Info

Description - optional Info

e.g. SSH for admin desktop

▼ Security group rule 3 (TCP, 443, 0.0.0.0/0) Remove

Type Info

HTTPS ▾

Protocol Info

TCP

Port range Info

443

Source type Info

Custom ▾

Source Info

Description - optional Info

e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. X

Add security group rule

Launch instance butonuna bas ve. Instances sayfasında running olana kadar bekle.

Instances (1) Info							Connect	Instance state ▼	Actions ▼
Find Instance by attribute or tag (case-sensitive)							All states ▼		
☐	Name ↗	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status			
☐	Nginx_Web	i-0ecbb285e70e2a66e	Running	t2.micro	Initializing	View alarms			

2 Elastic IP Atama

Bu sunucu Web server olarak hizmet edeceği için IP adresinin sabit kalması için Elastic IP ile kalıcı IP atanmasını sağlayalım:

Network & Security → Elastic IPs → Allocate Elastic IP

EC2 > Elastic IP addresses > Allocate Elastic IP address

Allocate Elastic IP address Info

Elastic IP address settings Info

Public IPv4 address pool

☒ Amazon's pool of IPv4 addresses

☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) Learn more

☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) Learn more

☐ Allocate using an IPv4 IPAM pool. (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group Info

us-east-1

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. Learn more

Create accelerator

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tag

Cancel

Allocate

Allocate butonuna tıklayarak static Public IP atanmasını sağlayalım.

Elastic IP addresses → Actions → Associate Elastic IP address

Elastic IP addresses (1/1) Info					Actions ▲	Allocate Elastic IP address
Find elastic IP addresses by attribute or tag						
Public IPv4 address : 34.202.44.116					Clear filters	
☒	Name	Allocated IPv4 address	Type	Allocation ID		
☒	-	34.202.44.116	Public IP	eipalloc-0fc28		

View details

Release Elastic IP addresses

Associate Elastic IP address

Disassociate Elastic IP address

Update reverse DNS

Enable transfers

Disable transfers

Accept transfers

Resource type: **instance**

Instance: **Nginx_Web**

EC2 > Elastic IP addresses > Associate Elastic IP address

Associate Elastic IP address [Info](#)

Choose the instance or network interface to associate to this Elastic IP address (34.202.44.116)

Elastic IP address: 34.202.44.116

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

⚠ If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance
i-0ecbb285e70e2a66e

Private IP address
The private IP address with which to associate the Elastic IP address.
Choose a private IP address

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
☐ Allow this Elastic IP address to be reassociated

Cancel Associate

Associate butonuna tıklayarak oluşturduğum Static IP **Nginx_Web** sunucumuza atanmasını sağlayalım.

Instances (1/1) [Info](#)

Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch](#)

Find Instance by attribute or tag (case-sensitive) All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability zone
☒	Nginx_Web	i-0ecbb285e70e2a66e	Running	t2.micro	2/2 checks passed	View alarms	us-east-1

i-0ecbb285e70e2a66e (Nginx_Web)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary [Info](#)

Instance ID i-0ecbb285e70e2a66e	Public IPv4 address 34.202.44.116 open address	Private IPv4 addresses 172.31.19.154
IPv6 address -	Instance state Running	Public DNS ec2-34-202-44-116.compute-1.amazonaws.com

Artık Elastic IP sunucumuza atanmış durumda. Sunucumuz kapatılıp tekrar açılrsa bile hep aynı IP ile başlayacak.

3 Sunucuya Bağlanma

Sunucuya ssh ile bağlanabilmek için daha önce oluşturduğumuz **ec2_ssh_key.pem** dosyasının izinlerini ayarlayalım.

```
chmod 400 ec2_ssh_key.pem
```

Şimdi bu sunucuya bağlanalım

```
ssh -i ec2_ssh_key.pem ec2-user@34.202.44.116
```

```
(base) Downloads ➡ chmod 400 ec2_ssh_key.pem  
(base) Downloads ➡ ssh -i ec2_ssh_key.pem ec2-user@34.202.44.116  
The authenticity of host '34.202.44.116 (34.202.44.116)' can't be established.  
ED25519 key fingerprint is SHA256:+WckY4Sx4eR65l6KJw9I9atXHL+m0RJQtHDJwp0iD0A.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '34.202.44.116' (ED25519) to the list of known hosts.
```

```
,      #_  
~\     #####_        Amazon Linux 2023  
~~~   \#####\  
~~~~  \###|  
~~~~    \#/  ____  https://aws.amazon.com/linux/amazon-linux-2023  
~~~~      V~' '->  
~~~~  
~~~~_. _./_____  
~~~~._/ _/  
~~~~_/m/'
```

```
[ec2-user@ip-172-31-19-154 ~]$
```

4 Nginx Kurulumu

```
sudo dnf install -y nginx
sudo systemctl enable --now nginx
systemctl status nginx
```

```
[ec2-user@ip-172-31-19-154 ~]$ systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Mon 2025-09-15 18:15:42 UTC; 55s ago
     Process: 4920 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 4950 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 4997 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 5033 (nginx)
     Tasks: 2 (limit: 1111)
   Memory: 2.5M
     CPU: 41ms
   CGroup: /system.slice/nginx.service
           └─5033 "nginx: master process /usr/sbin/nginx"
             └─5039 "nginx: worker process"
```

Nginx servisi hazır şimdi SSL sertifikası için gerekli olan paketleri kuralım.

5 Certbot (SSL) Kurulumu

```
sudo dnf install -y certbot python3-certbot-nginx
```

DNS Ayarları

Öncelikle Web sitemize domain adresi ile erişebilmesi için DNS kayıtlarında sunucumuzun IP adresini domain adresine tanımlamalıyız.

Domain adresi: devopsatolyesi.com ve www.devopsatolyesi.com

IP adresi 34.202.44.116

DNS kayıtları Cloudflare üzerinde olduğu için IP eklemesini Cloudflare DNS panelinde yapacağız.

devopsatolyesi.com DNS kaydı

DNS→records

Type: **A**

Name: **devopsatolyesi.com**

IPv4 address: **34.202.44.116**

Proxy status: **off**

TTL: **2 min**

The screenshot shows the Cloudflare DNS Records page. On the left sidebar, the 'DNS' menu is expanded, and 'Records' is selected. The main content area shows a table with one record for 'devopsatolyesi.com'. The record is of type 'A', has the IPv4 address '34.202.44.116', and the proxy status is 'off' (DNS only). The TTL is set to 'Auto'. Below the table, there is a 'Record Attributes' section with a 'Comment' field. The 'Save' button is highlighted with a red box.

Type	Name (required)	IPv4 address (required)	Proxy status	TTL
A	@devopsatolyesi.com	34.202.44.116	off (DNS only)	Auto

Record Attributes [Documentation](#)

The information provided here will not impact DNS record resolution and is only meant for your reference.

Comment

Enter your comment here (up to 100 characters).

Cancel Save

Save diyerek DNS kayıtlarımızın aktif olmasını sağlayabiliriz.

www.devopsatolyesi.com DNS kaydı

DNS→records

Type: **A**

Name: **www.devopsatolyesi.com**

IPv4 address: **34.202.44.116**

Proxy status: **off**

TTL: **2 min**

Analytics & logs

DNS

Records

Analytics

Settings

Email

SSL/TLS

www.devopsatolyesi.com points to 34.202.44.116.

Type	Name (required)	IPv4 address (required)	Proxy status	TTL
A	www	34.202.44.116	☒ Proxy status ☒ DNS only	2 min

Use @ for root

Record Attributes [Documentation](#)

The information provided here will not impact DNS record resolution and is only meant for your reference.

Save diyerek DNS kayıtlarımızın aktif olmasını sağlayabiliriz.

DNS kayıtlarının aktif olduğunu doğrula:

```
nslookup www.devopsatolyesi.com
nslookup devopsatolyesi.com
```

```
(base) ~ ➡ nslookup www.devopsatolyesi.com
Server:          1.1.1.1
Address:         1.1.1.1#53

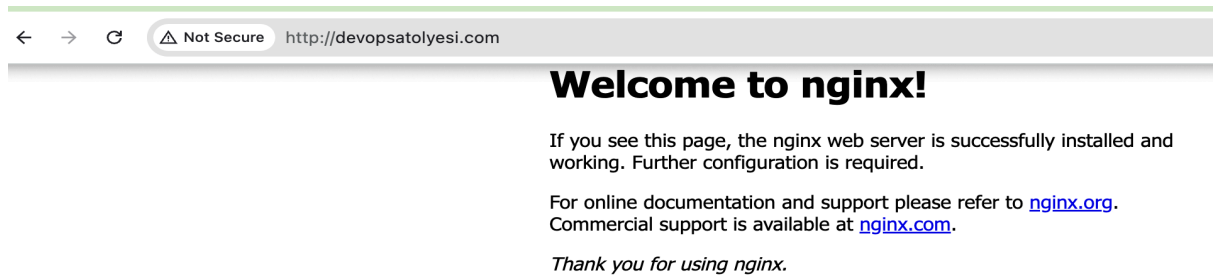
Non-authoritative answer:
Name:   www.devopsatolyesi.com
Address: 34.202.44.116

(base) ~ ➡ nslookup devopsatolyesi.com
Server:          1.1.1.1
Address:         1.1.1.1#53

Non-authoritative answer:
Name:   devopsatolyesi.com
Address: 34.202.44.116

(base) ~ ➡
```

Ayrıca Browserdan sitemize erişilip erişilmediğini test edelim.



Gördüğünüz gibi site geliyor ancak SSL sertifikası aktif değil.

6 Nginx Server Block (Host) Tanımlama

Certbot'un SSL ekleyebilmesi için `server_name` konfigürasyonu gerekli.

```
sudo nano /etc/nginx/conf.d/devopsatolyesi.conf
```

```
server {  
    listen 80;  
    server_name devopsatolyesi.com www.devopsatolyesi.com;  
  
    root /usr/share/nginx/html;  
    index index.html;  
  
    location / {  
        try_files $uri $uri/ =404;  
    }  
}
```

✓ Nginx test edip yeniden başlat:

```
sudo nginx -t  
sudo systemctl restart nginx
```

7 SSL Sertifikası Alma

Şimdi Web sitemize erişimi sağlayacak domain adresi için Let's encrypt sistesinden SSL sertifika alınması için aşağıdaki komutu çalıştırmalıyız.

```
sudo certbot --nginx -d devopsatolyesi.com -d www.devopsatolyesi.com
```

```
[ec2-user@ip-172-31-19-154 ~]$ sudo certbot --nginx -d devopsatolyesi.com -d www.devopsatolyesi.com
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Enter email address (used for urgent renewal and security notices)
(Enter 'c' to cancel): hkg, il, an, g... | ...

-----
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.5-February-24-2025.pdf. You must
agree in order to register with the ACME server. Do you agree?
-----
(Y)es/(N)o: yes

-----
Would you be willing, once your first certificate is successfully issued, to
share your email address with the Electronic Frontier Foundation, a founding
partner of the Let's Encrypt project and the non-profit organization that
develops Certbot? We'd like to send you email about our work encrypting the web,
EFF news, campaigns, and ways to support digital freedom.
-----
(Y)es/(N)o: no
Account registered.
Requesting a certificate for devopsatolyesi.com and www.devopsatolyesi.com

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/devopsatolyesi.com/fullchain.pem

Deploying certificate
Successfully deployed certificate for devopsatolyesi.com to /etc/nginx/conf.d/devopsatolyesi.conf
Successfully deployed certificate for www.devopsatolyesi.com to /etc/nginx/conf.d/devopsatolyesi.conf
Congratulations! You have successfully enabled HTTPS on https://devopsatolyesi.com and https://www.devopsatolyesi.com

-----
If you like Certbot, please consider supporting our work by:
* Donating to ISRG / Let's Encrypt: https://letsencrypt.org/donate
* Donating to EFF: https://eff.org/donate-le
-----
```

Email adresimizi vererek anlařmayı yes ile kabul edelim. Let's encrypt ve partnerlarından haber ve kampanya bilgilerini almak istemiyorsak no diyerek geçelim. Sertifika oluşturuldu. Browserdan görebiliriz.

→ **https://devopsatolyesi.com**

Certificate Viewer: devopsatolyesi.com

General Details

Issued To

Common Name (CN)	devopsatolyesi.com
Organization (O)	<Not Part Of Certificate>
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By

Common Name (CN)	E8
Organization (O)	Let's Encrypt
Organizational Unit (OU)	<Not Part Of Certificate>

Validity Period

Issued On	Monday, September 15, 2025 at 8:42:57 PM
Expires On	Sunday, December 14, 2025 at 8:42:56 PM

SHA-256 Fingerprints

Certificate	d649f7c3cfde34d32447feb75716fa9bd7231229146a6c158a9bc ae30883b184
Public Key	74efc158246189a7affed63b83a7dea71e6ded786fb99954697e9 26f179d76be

8 Sertifika Yenileme (Auto-Renew)

Let's Encrypt sertifikaları 90 gün geçerlidir. Otomatik yenileme için cron ayarlayalım.

Öncelikle Amazon Linux 2023'te **cronie** (cron servisi) paketi varsayılan olarak kurulu gelmiyor. Yani **crontab** komutu da bulunmuyor.

Bunu çözmek için şu adımları izle:

Cron Servisini Kur

```
sudo dnf install -y cronie
```

Servisi Etkinleştir ve Başlat

```
sudo systemctl enable --now crond
```

Durumu kontrol et:

```
systemctl status crond
```

✓ Active (running) görmelisin.

Crontab'a Görev Ekle

crontab -e

açılan editöre aşağıdaki komutu ekle

```
0 2 * * * root /usr/bin/certbot renew --quiet
```

Kaydet ve çık.

🧹 Cleanup

- EC2 terminate,
- EIP release